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Validity of muscle activation estimated with predicted ground reaction force in inverse dynamics based musculoskeletal simulation during gait

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ABSTRACT

The inverse dynamics based musculoskeletal simulation needs ground reaction forces (GRF) as an external force input. GRF can be predicted from kinematic data. However, the validity of estimated muscle activation using the predicted GRF has remained unclear. Therefore, the purpose of this study was to determine the validity of estimated muscle activation with predicted GRF in the inverse dynamics based musculoskeletal simulation. To perform musculoskeletal simulations, an open-source motion capture dataset that contains gait data from 50 healthy subjects was used. CusToM was used for the musculoskeletal simulations. Two sets of inverse dynamics and static optimization were performed, one used predicted GRF (PRED) and another used experimentally measured GRF (EXP). Pearson's correlation was calculated to evaluate the similarity between EMG and estimated muscle activations for both PRED and EXP. To compare PRED and EXP, paired t-tests were used to compare the trial-wise muscle activation similarity and residuals. Relationships between joint moments and residuals were also tested. The overall muscle activation similarity was comparable in PRED ($R = 0.477$) and EXP ($R = 0.475$). The residuals were 2–4 times higher in EXP compared to PRED ($P < 0.001$). The hip flexion–extension moment was correlated to sagittal plane residual moment ($R = 0.467$). The muscle activations estimated using predicted GRF were comparable to that with measured GRF in the inverse dynamics based musculoskeletal simulation. Prediction of GRF helps to perform musculoskeletal simulations where the force plates are not available.

1. Introduction

Musculoskeletal modeling is used to simulate neuromuscular control in various human movements (Delp and Loan, 2000; Dembia et al., 2020; Hatze, 1976; Falisse et al., 2019; Heinrich et al., 2022; Navacchia et al., 2019; Uchida et al., 2016). This information is useful to understand the neuromuscular control strategy for healthy and pathological movements (Afschrift et al., 2018; Falisse et al., 2020, 2019; Navacchia et al., 2019; Pitto et al., 2019). Commonly the kinematics and kinetics data for musculoskeletal modeling is obtained with a marker-based three dimensional motion capture system that includes force plates in a laboratory space (Sylvester et al., 2021). These devices are gold standard for the biomechanical measurements (Winter, 2009). For clinical assessment, however, not all the hospitals and clinics can operate the costly motion capture system with force plates. To address this problem, previous studies proposed approaches to predict ground reaction force (GRF) (Dijkstra and Gutierrez-Farewik, 2015; Fluit et al., 2014; Muller et al., 2020; Skals et al., 2017).

A previous study utilized a foot–ground contact model to predict GRF in forward dynamics based musculoskeletal simulations (Falisse et al., 2019; Koelewijn and van den Bogert, 2020). The contact model applies a force when it penetrates the ground plane. The reported estimated muscle activations were in agreement with measured electromyography (EMG) (Falisse et al., 2019). It indicated that muscle activations and forces could be successfully estimated without force plates. However, this approach has some difficulties in performing the simulation. It is an optimal control problem that includes a large number of states, control and constraints. Although the performance and robustness has been improved, it is time consuming (36 min for half gait cycle (Falisse et al., 2019)) and may not converge to find the reasonable solution (Anderson and Pandy, 2001; Dembia et al., 2020). In contrast, inverse dynamics based simulation, especially static optimization, is relatively easy and stable to find optimal results (Erdemir et al., 2007).

Musculoskeletal simulations have been performed with inverse dynamics based approaches in previous studies (De Groote et al., 2016; Mokhtarzadeh et al., 2013; Ueno et al., 2020; Zargham et al., 2019). A

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challenge in the inverse dynamics based simulation is the errors between measured kinematics and force data (Erdemir et al., 2007), which is called residual forces and moments. These residuals affect inverse dynamics results and subsequently the static optimization results, e.g. muscle activations and forces (Faber et al., 2018). A previous study proposed an approach to predict GRF from motion with contact models and an optimization technique (Muller et al., 2020). In this contact force optimization step, the sum of squared contact forces was minimized while respecting the multibody dynamics. Residuals were mostly eliminated in this step. They reported that the correlation coefficients and root mean square error averaged over the task between L5/S1 joint moment computed with predicted and measured GRF were 0.95 and 14 Nm in flexion/extension direction during a manual material handling task. With its accuracy with eliminated residual forces and moments, it seemed that their approach can derive accurate results from the static optimization in musculoskeletal modeling. However, the validity of muscle activation estimated with predicted GRF has not been reported. For gait trials, the validity of predicted GRF and joint moments using Muller's approach remain unclear as well. Predicted GRF in double leg stance phase of gait would be less accurate since objective function was simplified from Fluit et al. (2014) which addressed the indeterminacy problem of double leg stance phase. Therefore, the purpose of this study was to determine the validity of predicted GRF and joint moments as well as the muscle activation estimated with predicted GRF in the inverse dynamics based musculoskeletal simulation during gait. It was hypothesized that the predicted GRF and joint moments would be accurate and could derive estimated muscle activation comparable to that with measured GRF.

Methods.

1.1. Experimental data

An open-source motion capture dataset that contains gait data from 50 healthy subjects (24 women and 26 men, 37.0 ± 13.6 years, 1.74 ± 0.09 m, 71.0 ± 12.3 kg) was used as the experimental data (Schreiber and Moissenet, 2019). Briefly, participants performed walking at five different speeds: between 0 and 0.4 $m.s^{-1}$, between 0.4 and 0.8 $m.s^{-1}$, between 0.8 and 1.2 $m.s^{-1}$, self-selected spontaneous speed and self-selected fast speed. 52 marker trajectories were recorded with 10 camera optoelectronic system sampled at 100 Hz (OQUS4, Qualisys, Sweden). Two force plates sampled at 1500 Hz (OR6-5, AMTI, USA) were used to record 3D ground reaction forces and moments. Following SENIAM recommendation, a wireless EMG system sampled at 1500 Hz (Desktop DTS, Noraxon, USA) was used to record 8 muscles of the right leg: gluteus maximus, gluteus medius, rectus femoris, vastus medialis, semitendinosus, gastrocnemius medialis, soleus, and tibialis anterior. All these systems were synchronized using the Qualisys Track Manager software (QTM 2.8.1065, Qualisys, Sweden). The data in which the subject failed to step on the force plates with each foot was excluded from the analysis.

1.2. Data processing

C3D files that stored the pre-processed marker trajectories and force plate data distributed with the dataset were used for this study. According to Schreiber and Moissenet (2019), the marker trajectories were smoothed using a fourth order Butterworth low pass filter with a 6 Hz cut-off frequency, while the ground reaction forces and moments were smoothed using a second order Butterworth low pass filter with a 15 Hz cut-off frequency in their study. Raw EMG data that distributed with CSV files were band pass filtered at 10–400 Hz, full-wave rectified and low pass filtered at 6 Hz in this study according to a previous study (Navacchia et al., 2019). Since EMG signals during maximum voluntary contraction were not available from the dataset, EMG was normalized to the peak value of estimated muscle activations as similar to a previous study (Rajagopal et al., 2016). When the EMG was compared to PRED, it was normalized to the peak value of the estimated muscle activation from PRED in the trial, while EMG was normalized to the peak value of the estimated muscle activation from EXP when the EMG was compared to EXP.

1.3. Musculoskeletal modeling

A summary of the musculoskeletal simulation pipeline is shown in Fig. 1. CusToM, a MATLAB toolbox was used for the musculoskeletal simulations (Muller et al., 2019). Full body model that consists of 42 degrees of freedom, 82 lower extremity muscles and 28 contact points on the feet was used as a generic model. Subject models were calibrated by optimizing the segment length and marker placements of the generic model (Andersen et al., 2010). Then inverse kinematics, external force prediction, inverse dynamics and static optimization were performed. The external force prediction in CusToM was designed to perform the approach proposed by the previous study (Muller et al., 2020). It first detected the contact points that were in contact state with the ground. When the height and speed of a contact point were below 0.05 m and 0.8 m/s, respectively, the contact point was considered to be in contact with the ground. Then the contact points for each foot in contact state were optimized by minimizing sum of squared forces on the contact points. In the equality constraint of the optimization, the forces and moments around the pelvis that simply calculated from the contact force vectors and pelvis-contact points distance were set equal to the forces and moments at the pelvis calculated from inverse dynamics only with the kinematics without external forces. The maximum force for each contact point was limited to 40 % of the subject's body weight. The predicted GRF was used for the calculation of inverse dynamics. The joint kinematics and GRF data were processed with fourth order Butterworth lowpass filter at 6 Hz applied twice in both the forward and reverse directions before inverse dynamics. Inverse Dynamics derived the joint moments and residual forces and moments. Finally static optimization was performed to simulate muscle activations and forces. The sum of cubed muscle activations was minimized in static optimization as it was the default setting in CusToM. In this study, two sets of inverse dynamics and static optimization were performed, one used predicted GRF (PRED) and another used experimentally measured GRF (EXP).

1.4. Statistical analysis

To evaluate the validity, root mean square error (RMSE) and Pearson's correlation coefficient for all time frames of GRF and joint moments between PRED and EXP were evaluated. Force and moment values were normalized to the subject's bodyweight (BW) and bodyweight times body height (BW*BH), respectively. Pearson's correlations between EMG and estimated muscle activations for both PRED and EXP were calculated to evaluate the validity using all time frames as samples for the analysis as similar to a previous study (Zargham et al., 2019). To evaluate the differences between PRED and EXP, paired *t*-test was used for the trial-wise Pearson's correlation coefficient of the estimated muscle activation, as well as the root mean square of the residual forces and moments. To investigate how residuals relate to computation of the joint moments, Pearson's correlation coefficient between residuals and joint moments using all time frames were computed. The values for residuals and joint moments were calculated as the absolute value of the PRED-EXP difference. Stance phase for GRF, full right gait cycle for joint moments and muscle activation were included for statistics. With regard to the residuals, the loading response phase (first double leg stance phase) lacked the GRF measurement on the other foot and was excluded from the statistical analysis. Note that the second double leg stance phase (pre-swing phase) was included in the analysis. The figures show full gait cycle.

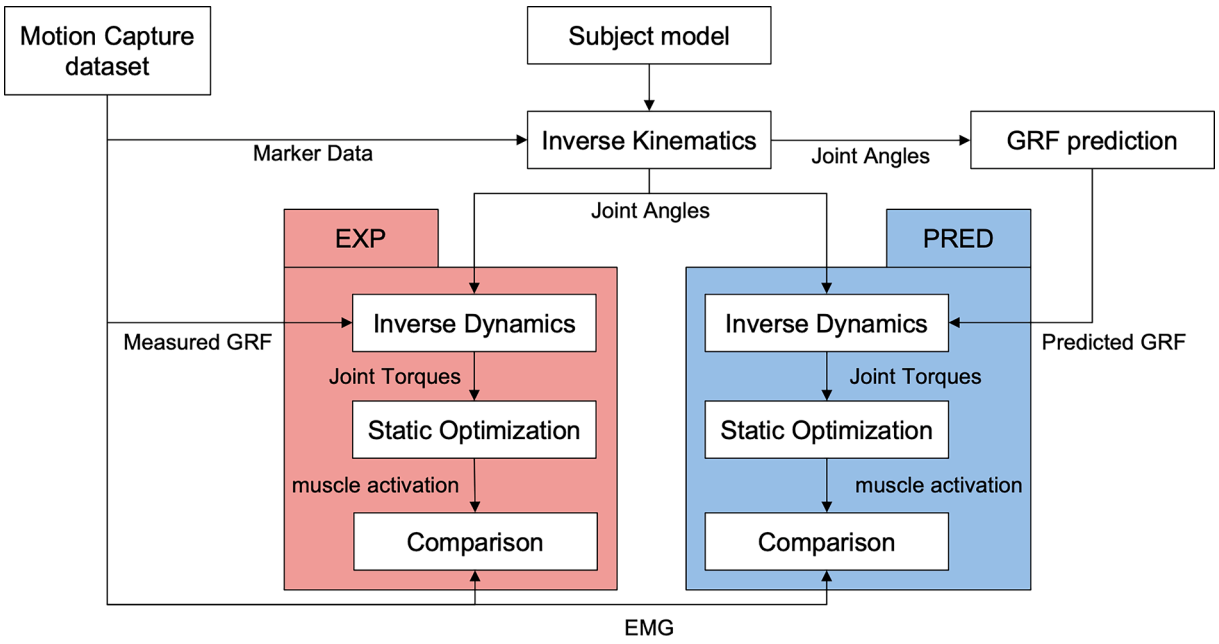


Fig. 1. A summary of the musculoskeletal simulation pipeline.

2. Results

RMSE between PRED and EXP in GRF and ground reaction moment (GRM) were 0.021–0.084 BW and 0.009–0.034 BW*BH (Fig. 2, Table1). Pearson's correlation coefficient between PRED and EXP in GRF and GRM were 0.750–0.983 (Fig. 2, Table1). The RMSE between PRED and EXP in joint moments were 0.002–0.023 BW*BH (Fig. 3, Table2). Pearson's correlation coefficient between PRED and EXP in joint moments were 0.794–0.933 (Fig. 3, Table2). The mean value of the correlation coefficient between EMG and estimated muscle activations from the 8 muscles were $R = 0.477$ ($P < 0.001$) in PRED and $R = 0.475$ ($P < 0.001$) in EXP (Fig. 4, Table 3). The paired t-test did not detect a significant difference in correlation coefficient between PRED and EXP ($P = 0.087$). The residuals were 2–4 times higher in EXP compared to PRED ($P < 0.001$, Fig. 5, Table 4). The highest correlation coefficient between residuals and joint moments was found between the hip

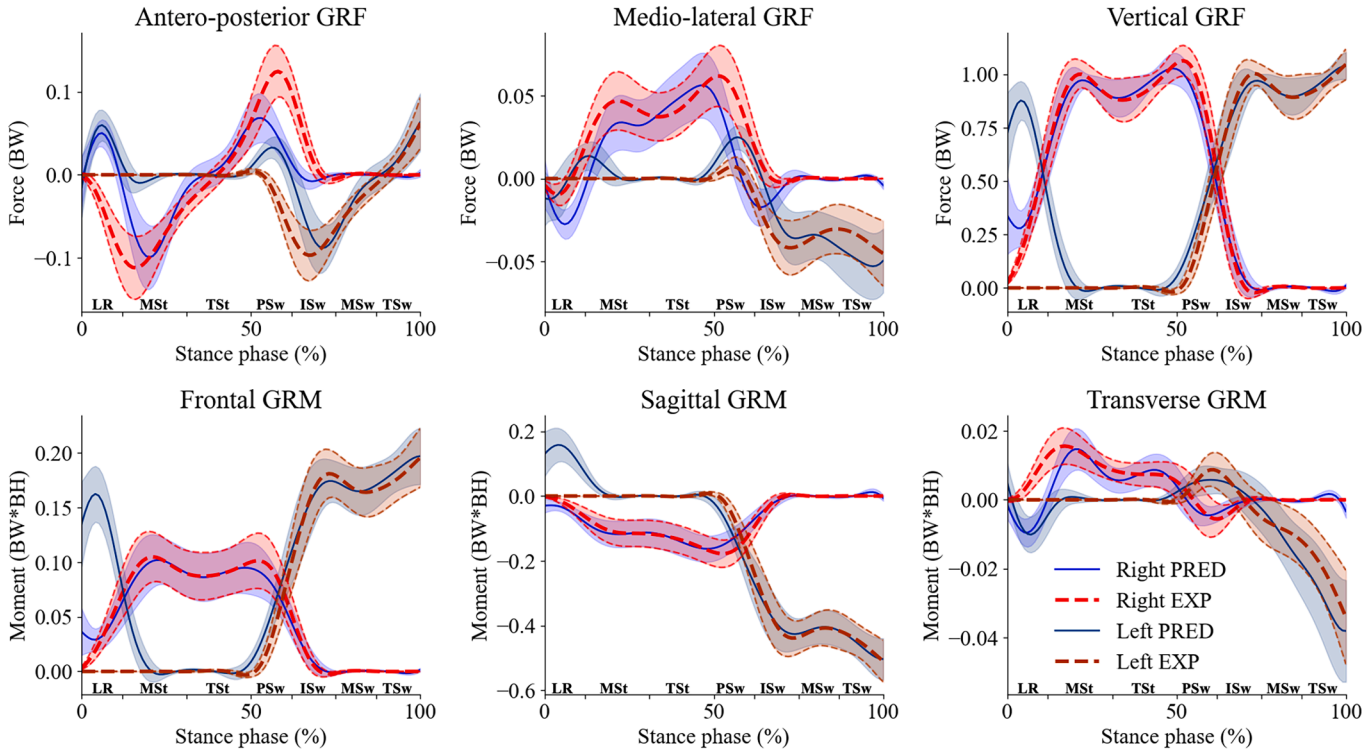


Fig. 2. Comparison of the mean and standard deviation curves for the ground reaction forces and moments between PRED (blue: right, dark blue: left) and EXP (red: right, dark red: left). The values are expressed in the global coordinate system. The figure was generated only with the data in which the first step was the right foot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Root mean square error (RMSE) and Pearson's correlation coefficient R for ground reaction forces and moments between PRED and EXP.

	RMSE		R	
	Left	Right	Left	Right
Antero-posterior GRF	0.051	0.051	0.765	0.768
Medio-lateral GRF	0.021	0.021	0.823	0.833
Vertical GRF	0.083	0.084	0.969	0.968
Frontal GRM	0.013	0.013	0.975	0.975
Sagittal GRM	0.034	0.031	0.982	0.983
Transverse GRM	0.012	0.009	0.750	0.819

RMSE of GRFs and GRMs are in bodyweight (BW) and bodyweight times body height (BW*BH), respectively.

flexion–extension moment and sagittal plane residual moment (R = 0.458, Table 5).

3. Discussion

The purpose of this study was to determine the validity of predicted GRF, joint moments and muscle activations estimated with predicted GRF in the inverse dynamics based musculoskeletal simulation during gait. Although some errors existed in GRF and joint moments, Fig. 4 shows that the muscle activations estimated using predicted GRF were in agreement with EMG. The overall correlation coefficient between EMG and estimated muscle activations using predicted GRF from the 8 muscles were comparable to the estimated muscle activations using measured GRF. These results indicated that the predicted GRF can be used to estimate muscle activations similarly to when the measured GRF is used in inverse dynamics based musculoskeletal simulation during gait. Despite the comparable correlation coefficients, the magnitude of estimated muscle activations as well as the muscle forces were different between PRED and EXP and not validated against EMG. This point might limit the clinical application where the magnitude of muscle forces or joint reaction forces are important.

The waveforms of GRF and joint moments were similar between PRED and EXP for most of the phases (Fig. 2, Fig. 3). However, large errors in anterior-posterior and medio-lateral GRF were found during double leg stance phase (around 0–15 % and 50–65 %). This would be because of the limitation of the Muller's approach used in this study. The horizontal GRF of both legs might balance in the opposite direction during double leg stance phase. The optimization that minimized only the sum of squared contact forces would not be the best approach for gait trials since underestimated contact forces could still satisfy the multi-body dynamics. Fluit et al. (2014) addressed this problem by minimizing weighted sum of squared muscle activations and contact forces simultaneously. It may estimate larger horizontal GRF if it lowers the joint moments to minimize muscle activation. This aspect should be included in the optimization especially for gait trials.

The obtained correlation coefficient between estimated activations and EMG in this study were similar to a previous report that performed inverse dynamics based simulations (Zargham et al., 2019). This is novel findings that the muscle activations may be estimated with predicted GRF comparably to the measured GRF in some scenarios. This could be explained with the higher residuals in EXP. When combined with the kinematics, which would be distorted during marker trajectories

Table 2
Root mean square error (RMSE) and Pearson's correlation coefficient R for joint moments between PRED and EXP.

	RMSE	R
Hip flexion	0.030	0.803
Hip adduction	0.010	0.889
Hip internal rotation	0.003	0.736
Knee extension	0.022	0.466
Ankle dorsi flexion	0.015	0.843

RMSE is in bodyweight times body height (BW*BH).

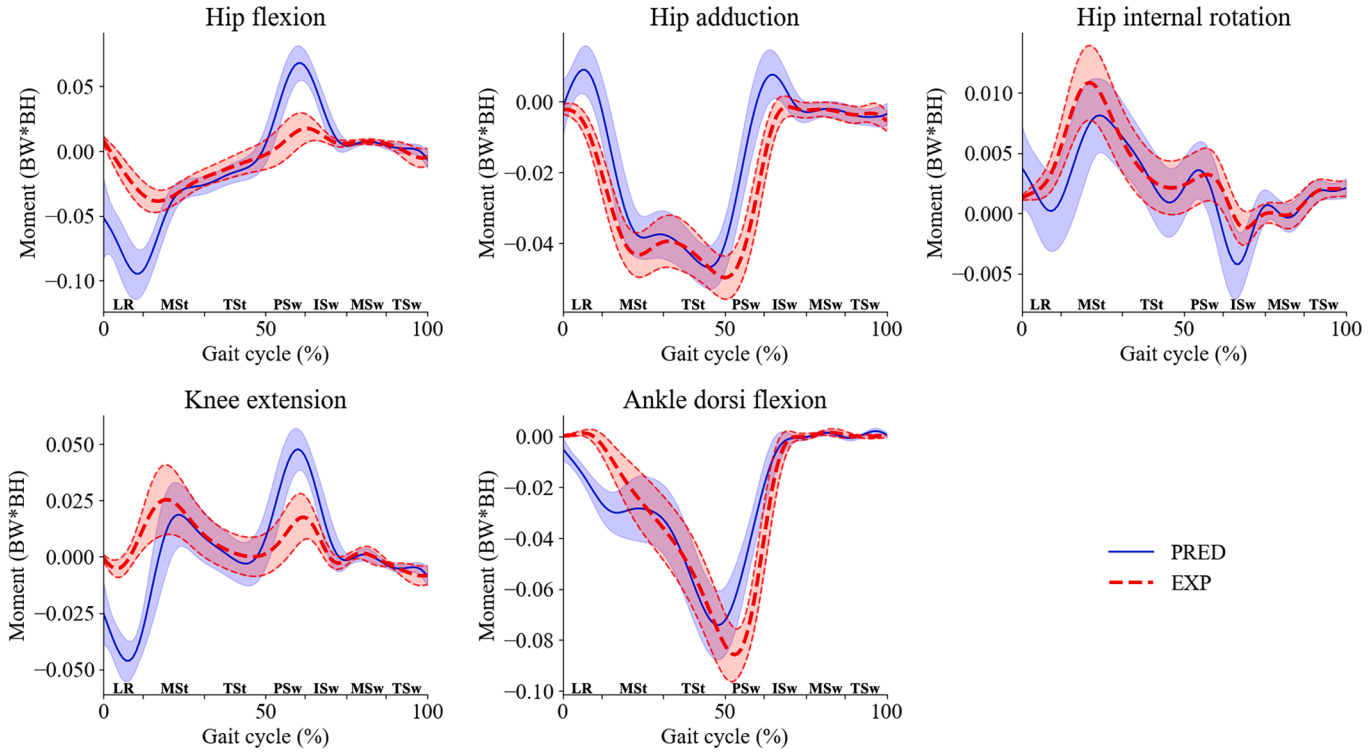


Fig. 3. Comparison of the mean and standard deviation curves for the joint moments between PRED (blue) and EXP (red). The figure was generated only with the data in which the first step was the right foot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

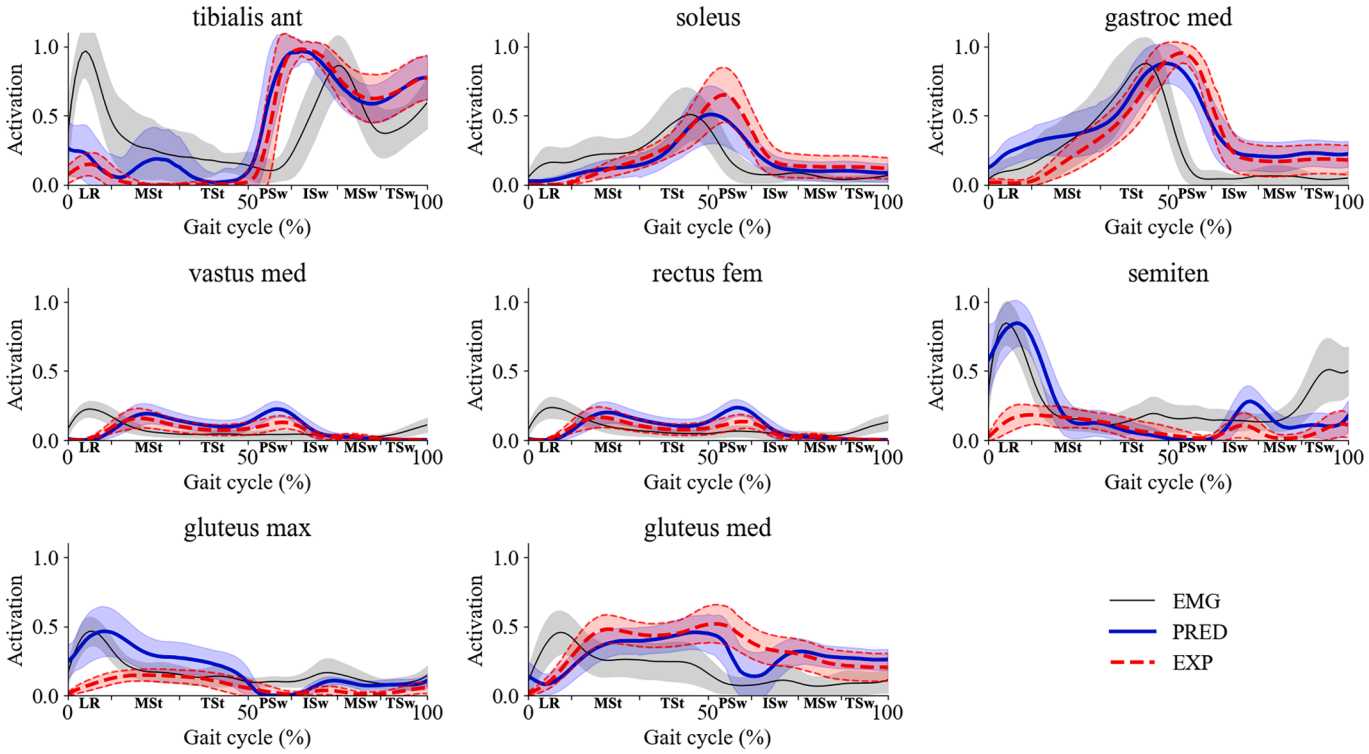


Fig. 4. Estimated muscle activations using predicted GRF (PRED) and measured GRF (EXP) compared to EMG. The figure was generated only with the data in which the first step was the right foot.

Table 3
Pearson's correlation between EMG and estimated muscle activation when predicted (PRED) or measured (EXP) GRF was used. The *P*-values are from the paired *t*-test that compared trial-wise *R*-values between PRED and EXP.

	PRED	EXP	*t-test P-value*
tibialis anterior	0.239	0.327	< 0.001
soleus	0.547	0.512	0.872
gastrocnemius medialis	0.617	0.577	0.002
vastus medialis	− 0.137	0.028	< 0.001
rectus femoris	− 0.048	0.109	< 0.001
semitendinosus	0.431	0.217	< 0.001
gluteus maximus	0.533	0.311	< 0.001
gluteus medius	0.173	0.270	< 0.001
all	0.477	0.475	0.087

processing, model scaling and inverse kinematics (mean marker error in inverse kinematics was 1.14 cm), the mismatch between the kinematics and the measured GRF might derive higher error in multibody dynamics and affected the estimation of muscle activation in EXP (Erdemir et al., 2007; Faber et al., 2018; Hicks et al., 2015). The residuals should be minimized with careful experimental measurement and possibly the optimization technique used in a previous study when force plate data is available (Faber et al., 2018; Hicks et al., 2015). A synthetic data with artificial errors in kinematic and GRF may be useful to study how residuals affect estimated muscle activations. The findings indicated that the muscle activations can be estimated without force plates. It encourages the future works of the musculoskeletal simulations where the force plates are not available but kinematics can be measured.

The highest correlation coefficient between residuals and joint moments was found between the hip extension moment and sagittal plane residual moment (Table 3). This result was similar to a previous study that reported that the larger alteration was found in hip joint moment after removing residuals (Faber et al., 2018). This would explain the lower correlation coefficient from the hip extensors such as semitendinosus and gluteus maximus in EXP. The correlation coefficient for quadriceps were very low in this study. This result was similar to the previous study that reported the low correlation coefficient of rectus femoris (- 0.175) (Zargham et al., 2019). The correlation coefficient for vastus medialis was not found in previous studies. Nevertheless, the quadriceps activations were very low during walking and might not be appropriate to assess quadriceps. The accuracy and validation of quadriceps activations should be tested in the different tasks that quadriceps works dominantly in a future study.

Some limitations should be noted for this study. First, the Muller's approach would not be the best solution for the double leg stance phase compared to the Fluit's approach as discussed above. Future study should implement better optimization to solve the indeterminacy problem of double leg stance phase. Second, only the walking of healthy subjects was investigated in this study. Differences between pathological and healthy subjects or changes after a treatment in muscle activation should be validated with a dataset that includes pathological cases for a better clinical application. For quadriceps activation, squatting or jumping tasks are better to be included in the dataset as well. Third, the results were not compared to the other approaches. Previous studies utilized approaches such as Residual Reduction Algorithm (RRA) that reduce residuals by altering the joint kinematics and center of mass location of torso in the optimization (Delp et al., 2007; Faber et al., 2018; Remy and Thelen, 2009). It remains unclear whether the measured kinematics with predicted GRF estimates the muscle activations than modified kinematics from RRA with measured GRF. The waveforms of muscle activations in forward dynamics based dynamic optimizations in a previous study (Falisse et al., 2019) seem to match EMG than this study. More physiologic dynamics should be considered to improve the estimation of muscle activations. Finally, the predictions of GRF are performed only with optical motion capture data. The approach to predict GRF used in this study is dependent on the kinematics data. The validity of predicted GRF on IMU based or computer vision based motion capture were not tested. This aspect should be tested when those procedures are combined.

The muscle activations estimated with predicted GRF was

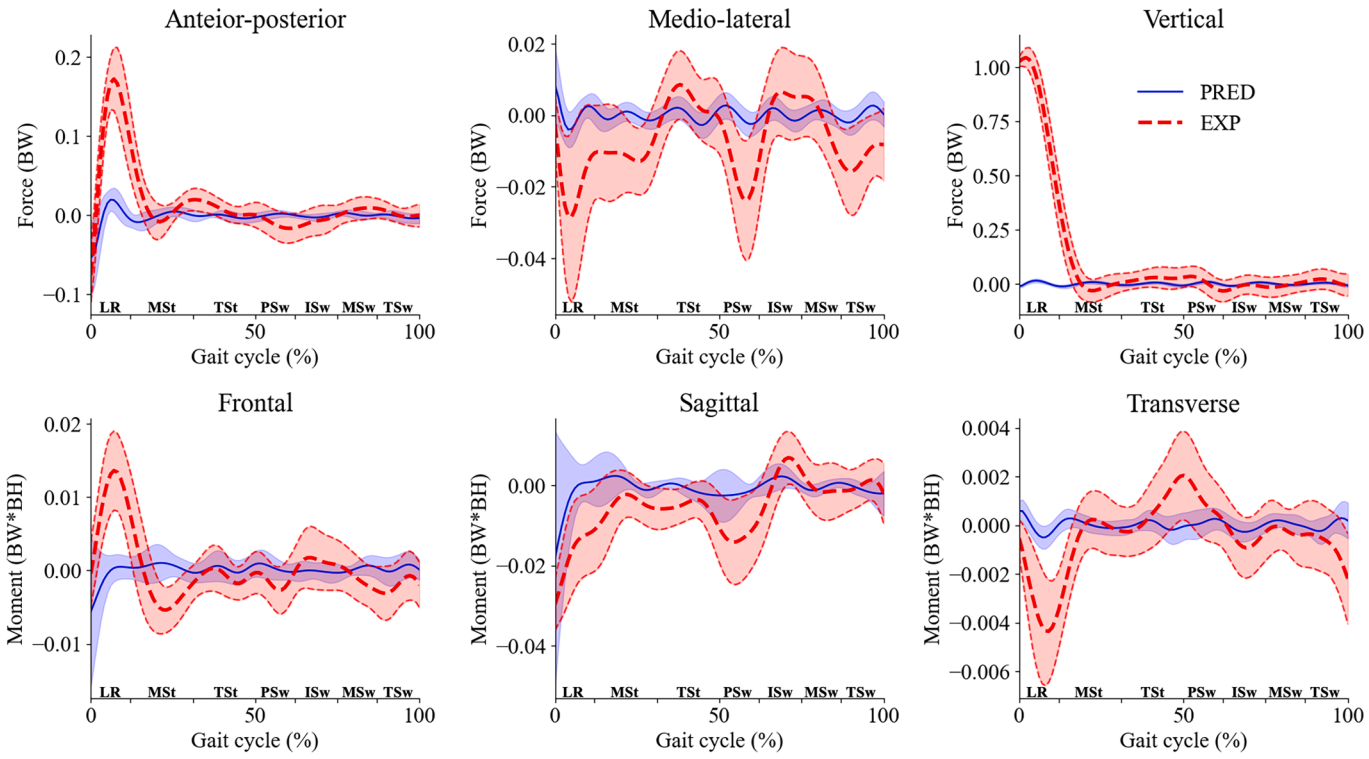


Fig. 5. Comparison of the mean waveforms of residuals between PRED (blue) and EXP (red). The figure was generated only with the data in which the first step was the right foot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Comparison of the root mean square of residual forces and moments between PRED and EXP and *P*-values from the trial-wise paired *t*-test.

Residual forces and moments	PRED	EXP	*t-test P-value*
Antero-posterior force	0.009	0.039	< 0.001
Medio-lateral force	0.008	0.022	< 0.001
Vertical	0.007	0.018	< 0.001
Frontal moment	0.002	0.006	< 0.001
Sagital moment	0.002	0.006	< 0.001
Transverse moment	0.002	0.006	< 0.001

Root mean square of residuals are in bodyweight times body height (BW*BH).

Table 5
Correlation coefficients between PRED-EXP difference in residuals and PRED-EXP difference in the right leg joint moments.

Residuals	Hip flex-ext	Hip add-abd	Hip in-ext rot	Knee ext-flex	Ankle DF-PF
Antero-posterior force	0.162	0.088	0.159	0.151	0.149
Medio-lateral force	0.188	0.188	0.195	0.182	0.173
Vertical force	0.044	0.136	0.102	0.077	0.105
Frontal moment	0.000	0.058	0.139	0.019	− 0.017
Sagital moment	0.458	0.306	0.240	0.417	0.359
Transverse moment	0.058	0.032	0.067	0.045	0.001

comparable to that with measured GRF in inverse dynamics based musculoskeletal simulation. Residuals affect the estimation of muscle activations. The residual moment in the sagittal plane is related to lowering the quality of the estimated muscle activations of hip extensors. Prediction of GRF helps to perform musculoskeletal simulations where the force plates are not available.

CRediT authorship contribution statement

Ryo Ueno: Writing – original draft, Software, Methodology, Conceptualization. **Yasuaki Tsuyuki:** Writing – review & editing, Investigation. **Harukazu Tohyama:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ryo Ueno and Yasuaki Tsuyuki are employee of ORGO Inc.

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